Prediction of Student's Performance Using Modified KNN Classifiers

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Abstract The Educational field research using Data Mining techniques increased rapidly. The aim of Data Mining on Educational data is to discover the hidden patterns and knowledge about student's performance. This study focus on predicting the student's performance according to their marks through classification using modified KNN classifiers such as Cosine KNN, Cubic KNN, and Weighted KNN. The Dataset of the students of the 11th Grade of scientific branch at Gaza Strip secondary schools contains 13 parameters, 11 parameters of the subject's marks, average parameter and Grade parameter. The classifiers should predict the performance (Grade) the student will gain depending on the marks of two subjects. The early prediction of the student's grade can help the principals to take decisions in order to help schools to initially identify students with low academic achievement and find ways to support them.

Keywords
NN techniques, KNN classification, Weighted KNN, Cosine KNN, Cubic KNN, Student's performance, Data Mining, Educational research

1.0 INTRODUCTION

The using of information technology in various fields has led the large volumes of data storage in various formats like records, files, documents, images, scientific data and many new data formats. The data collected from different applications require a proper method of extracting knowledge from large repositories for better decision making [1]. Knowledge discovery in databases (KDD) is used for discovery of useful information from large collections of data. The data mining applies various methods and algorithms in order to discover and extract patterns of stored data. Data mining techniques have been introduced into new fields of Statistics, Machine Learning, Pattern Reorganization, Artificial Intelligence and computation capabilities. Educational Data Mining (EDM) uses many techniques such as Decision Trees, Neural Networks, Naïve Bayes, K- Nearest neighbor, and many others. These techniques are used to discover knowledge, such as classifications, association rules and clustering. The discovered knowledge can be used for prediction regarding enrollment of students in a particular course, prediction about students' performance and so on. Nearest Neighbor classification was proposed by P. E. Hart and T. M. Cover in 1966 – 67, and now it is very popular in application and research field (Papadopoulos 2006). The reason of its popularity is the simplicity and efficiency in programming. The searching result needs high memory and computation requirements, but after implementing NN technique, it enhanced and reduced the usage of memory and computation time. The NN classifier idea is to classify X find its closer neighbor among the training points (call it X') and assign to X the label of X'. The K Nearest Neighbors rules are a well-known classifiers in Data Mining and Machine learning tasks(Kononenko and Kukar 2007). It is one of the most effective and useful algorithms in DM(Papadopoulos 2006), Pattern recognition(Shakhnarovich, Indyk et al. 2006), and considered as one of top 10 methods in DM(Wu, Kumar et al. 2008). As shown in Fig.1, the KNN classifier finds the K training samples x_r, r= 1… k which are closest in distance to x, and then classify them using majority vote among the k neighbors. The amount of computation can be intense when the training data is large. This is because the distance between a new data point and every training point has to be computed and sorted.

To adjust the complexity of KNN, we use the number of neighbors k, which is the only parameter used to determine the complexity. The larger k is, the smoother the classif classification boundary becomes. In other word, the complexity of KNN becomes lower when k increases.

Input: D, the set of k training objects, and test object $z = (\mathbf{x}', y')$

Process:

Compute $d(\mathbf{x}', \mathbf{x})$, the distance between z and every object, $(\mathbf{x}, y) \in D$.

Select $D_z \subseteq D$, the set of k closest training objects to z.

Output: $y' = \underset{v}{\operatorname{argmax}} \sum_{(\mathbf{x}_i, y_i) \in D_z} I(v = y_i)$

Fig.6: KNN classifier

KNN classification performance can be improved through supervised learning. The Nearest prototype classifier is a classification model that assigns to observations the class labels of training samples whose mean is closest to the observation. In this study, we compare different types of NN classifiers that derived from the basic KNN classifier. These classifiers differ in the distance metric used and the amount of neighbors, which is the K value. Dealing with raw data is very complex to extract any useful information or knowledge. That is the reason why the data science has established for, and therefore using customized algorithms to work on the data are helpful.

According to these reasons, we want to classify the dataset of the 11th grade of the scientific branch of secondary schools in Gaza Strip to find out the classification structure, and show the relations between the item sets of the data. This can help in predicting the student's grade early and help schools to initially identify students with low academic achievement and find ways to support them.

2.0 Background and Related work

Classification techniques as KNN use different types of distance metrics which are used by applying the classification model. The following metrics are considered in this study.

The Euclidean distance uses Minkowski metric as a special case.

$$d2st = (xs - yt)(xs - yt)' \qquad \textbf{(1)}$$

For the special case of p = 1, the Minkowski metric gives the city block metric, for the special case of p = 2, the Minkowski metric gives the Euclidean distance, and for the special case of p = ∞, the Minkowski metric gives the Chebychev distance.

$$\mathbf{dst} = \sqrt[p]{\sum_{j=1}^{n} |xsj - ytj| \, p} \qquad \textbf{(2)}$$

Cosine distance is a measure of similarity between two none zero vectors of an inner product space that measures the cosine of the angle between them.

$$dst = \left(1 - \frac{(xsx't)}{\sqrt{(xsx's)(yty't)}}\right) \qquad \textbf{(3)}$$

A list of KNN Classifiers and its specification, which are used in this study, are explained as follows:

- Fine KNN Classifier: A classifier that makes finely detailed distinctions between classes with the number of neighbors set to 1.
- Medium KNN Classifier: A classifier that makes fewer distinctions than a Fine KNN, with the number of neighbors set to 10.
- Coarse KNN Classifier: A classifier that makes coarse distinctions between classes, with the number of neighbors set to 100.
- Cosine KNN Classifier: A classifier that uses the cosine distance metric.
- Cubic KNN Classifier: A classifier that uses the cubic distance metric.
- Weighted KNN Classifier: A classifier that uses distance weighting.

The rest of this section review previous studies which are related to our study in predicting student's performance, and review other studies which deals with KNN classification.

Fadhilah Ahmad et al. (Ahmad, Ismail et al. 2015) used classification methods in data mining to predict the student's academic performance in the first year of study at university in Computer science course. After applying the classification methods on the dataset, the result shows that Rule Based is the better model among the other techniques by receiving the highest accuracy value of 71.3%.

Other researchers, E. Osmanbegović and M. Suljić (Osmanbegović and Suljić 2012), used the surveys conducted during the summer semester at the university of Tazla, faculty of economics, among the first year students and data taken during the enrollment. After applying a three supervised data mining algorithms, they indicate that the Naive Bayes classifier outperforms decision tree and neural network methods in term of prediction accuracy.

S. K. Yadav et al. (Yadav, Bharadwaj et al. 2012) gives a study based on the decision tree supervised algorithm to predict the student performance based on the past performance data, and it helps identifying the dropouts and the students who need special attention.

Recently, A. Deshpande et al. (Deshpande, Pimpare et al. 2016) used data mining techniques to analyze, visualize and predict the performance of the students in tests, extra-curricular activities, sports to grade students for campus placement activity. They used ID3 decision tree algorithm to predict the performance of the students in end semester exam where previous and current year performance have been used in prediction. They also reported that the evaluation helps to know strong & weak areas of students and give more guidance on weak areas.

B. K. Baradwaj and S. Pal (Baradwaj and Pal 2012) use data mining to analyze student's performance. They use classification tasks to evaluate student's performance where the decision tree is used as a classification method. Knowledge that describes student's performance in the end semester examination has been extracted.

Dealing with varying types of algorithms needs to be considered by reviewing the previous studies and take a part of these algorithms and make an evaluation of their benefits, efficiency and effectiveness of the data classification.

To compare between NN searches algorithms, some researchers define metric and non metric spaces. The metric spaces used in NN search problems. Diverse distance formulas which are used in this problem to find the nearest data point are described. Then, different structures are introduced. These structures are used for indexing points and making the searching operation faster. The NN searching algorithms was used to find the nearest data point, and then different structures of NN searching are introduced (Mahapatra and Chakraborty 2015). These structures are used for indexing points and making the searching operation faster. Some of these structures as Ball-Tree, LSH, and KD-Tree are considered as a technique for NNS problem (Mahapatra and Chakraborty 2015).

The fast KNN classification algorithm has been used by the authors to explore the mean of the vector and the propose of this algorithm that alters the search order of the technique in the wavelet domain(Jeng-Shyang, Yu-Long et al. 2004).

Multi-label classification algorithm was implemented, which derived from the traditional KNN algorithm. According to the label sets of these neighboring instances, maximum a posteriori principle is utilized to determine the label set for the new instance(Zhang and Zhou 2005).

On the other point of view, varieties of NN techniques have to be considered. Comparing between the techniques of NN gives as two methods of comparison; structure less and structure base. These techniques have improvements over the basic KNN techniques. The KNN lies in the first category, where the data are classified into training data and sample data point. The distance evaluated from training to sample points, and the point with the lowest distance called nearest neighbor. The Second category of NN techniques is based on structures of data, for example principal axis Tree (PAT), Ball Tree, k-d Tree, orthogonal structure Tree (OST), Nearest feature line (NFL), Center Line (CL) etc. A ball tree for example is a binary tree and constructed using top down approach. This technique is an improvement over KNN in terms of speed. The improvements proposed to gain speed and space efficiency(Bhatia 2010).

The Fuzzy KNN basis algorithm is to assign membership as a function of the object's distance from its KNN and the membership in the possible classes. A sample of this kind of algorithms is JFKNN which is an improved version of the standard KNN. It is based on a learning scheme of class memberships, which provide each training instance with membership array which defines its fuzzy membership to each class. After the learning process, the final classification is performed similar to KNN (Jóźwik 1983).

For pattern classification, Cover, T. and Hart, P. (Cover and Hart 1967) used the decision rules to implement a method to classify a set of unclassified data from a previously classified points. For high dimensional data used in computer vision and machine learning, several types of algorithms were compared to get the best algorithm to deal with searching in high dimensional spaces (Muja and Lowe 2014).

Asif, R. et al (Asif, Merceron et al. 2014) did a study to help both of the teachers and the students to find out innovative ways using data mining techniques. They present a case study to predict the student's performance at an early stage of a university degree program, in order to help universities to initially identify students with low academic achievement and find ways to support them. Their results show that it is possible to predict the graduation performance in 4th year at university using only pre-university marks and marks of 1st and 2nd year courses.

Due to the previous researches, we find out that the researchers used several types of classification methods, but little studies use the modified KNN classifiers. In this study, we use the modified KNN classifiers for the prediction of the student's performance. The modified classifiers consist of 6 different types of KNN classifiers that differ from the original KNN classifier from the perspective of the Distance metrics and the amount of neighbors.

3.0 Methodology

As mentioned in the introduction, the aim of this study is to classify the dataset of the 11th grade of the scientific branch of secondary schools in Gaza Strip to find out the classification structure, and to show the relations between the item sets of the data. The study tries to utilize the classification models to predict the performance of the students according to their marks in two subjects. The classification model of KNN and its modified versions can predict the performance by comparing the input marks with the training dataset and find the nearest value of these marks which called the nearest neighbor.

MATLAB 2016 environment has been used to conduct all experiments. The students' marks dataset have been used to evaluate KNN classification algorithms. The dataset contains 13 attributes; 11 attributes contain the marks of subjects, Average and Grade. The dataset was collected from the database of ministry of education in Gaza strip. Classification learner of Matlab 2016 is used to apply the different KNN classification algorithms on the dataset. Fig.2 shows sample examples of students' marks for 14 students.

Depending on two subject's marks, the proposed classification model uses the dataset in order to predict the student's performance

religon	Arabic	English	Math	Physics	Chemistry	Biology	IT	Management	Sport	Art	Average	Grade
96	129	97	87	61	68	38	84	75	90	90	70.38	Good
90	111	85	115	81	79	66	84	81	90	91	74.85	Good
84	108	89	108	64	68	53	86	66	92	90	69.85	Good
93	115	132	160	80	89	61	89	86	92	91	83.69	Very Good
100	137	120	157	84	93	80	89	92	92	92	87.38	Very Good
69	75	55	90	57	51	31	61	51	92	91	55.62	Fair
94	108	76	131	60	66	61	78	79	92	92	72.08	Good
94	136	109	155	85	84	82	88	91	94	89	85.15	Very Good
100	146	147	193	100	100	98	98	98	94	90	97.23	Excelent
94	141	147	174	85	85	78	94	92	94	89	90.23	Excelent
87	131	115	158	78	80	80	93	86	94	88	83.85	Very Good
89	139	109	152	74	81	70	69	64	60	85	76.31	Good
84	133	111	153	79	83	69	82	72	60	84	77.69	Good
99	139	136	181	93	92	96	96	93	82	86	91.77	Excelent

Fig.7: Dataset

4.0 Results and discussion

In this section, the experimental evaluation aspects of classification methods using modified KNN classifiers such as Cosine KNN, Cubic KNN, and Weighted KNN are reported.

All experiments were implemented with MATLAB 2016 on a workstation with Core I7 CPU and 6 GB RAM. The dataset were randomly divided into two sets; 70% of the records have been used for training and the remaining 30% for testing the classifiers.

The experimental results of classification algorithms are shown in confusion matrices of the algorithm and prediction graphs. The model classify the data based on the grade of the students which are 6 categories; Excellent, Very Good, Good, Fair, Pass, and Fail.

Every classification algorithm used in this comparison has its own distance metric and value. Fine KNN uses Euclidian distance metric, Equal distance weight and default number of neighbors 1. The Accuracy of this algorithm based on the default values is 90.7%.
Regarding Medium KNN, it uses Euclidian distance metric, Equal distance weight and default number of neighbors 10. The accuracy of this algorithm based on the default values is 93.8%. Coarse KNN uses Euclidian distance metric, Equal distance weight and default number of neighbors 100. The Accuracy of this algorithm based on the default values is 92.3%. Cosine KNN uses cosine distance metric, Equal distance weight and default number of neighbors 10. The accuracy of this algorithm based on the default values is 81.3%. Cubic KNN uses Minkowski distance metric, Equal distance weight and default number of neighbors 10. The accuracy of this algorithm based on the default values is 93.5%. Weighted KNN uses Euclidian distance metric, Squared inverse distance weight and default number of neighbors 10. The Accuracy of this algorithm based on the default values is 94.1%.
The classification results according to the confusion matrix graph will show form Fig.3 to Fig.8, and in Fig.9 we show a sample of the prediction graph for all types of classifiers used in this study.

Table 1: Comparing Results

Classifier	Distance	neighbor	Accuracy	Time
Fine	Euclidian	1	90.7%	1.8972 Sec
Medium	Euclidian	10	93.8%	1.0067 Sec
Coarse	Euclidian	100	92.3%	1.3072 Sec
Cosine	Cosine	10	81.3%	1.0665 Sec
Cubic	Minkowski	10	93.5%	30.441 Sec
Weighted	Euclidian	10	94.1%	0.97885 Sec

Applying the KNN modified algorithms on the dataset as in Table 1 shows a variety of results in accuracy and the time it takes. It depends on the distance matrix applied by the model and the number of neighbors.
Fine KNN confusion matrix as in Fig.3 represents the prediction of classes according to the marks provided in by the dataset. It shows that the best prediction value is in excellent class and the worst prediction value in Fair class. The high amount of false prediction is in Fair class due to the confusion matrix.
The confusion matrix of Medium KNN shown in Fig.4 gives a better prediction values rather than the Fine KNN. It represents a best prediction of Fail class with 100% and the worse prediction of Pass class with 88%.
The best prediction of Coarse KNN model shown in Fig.5 of the confusion matrix is 95% of the Excellent class and the worse prediction is in Fail class with false prediction of 39%.

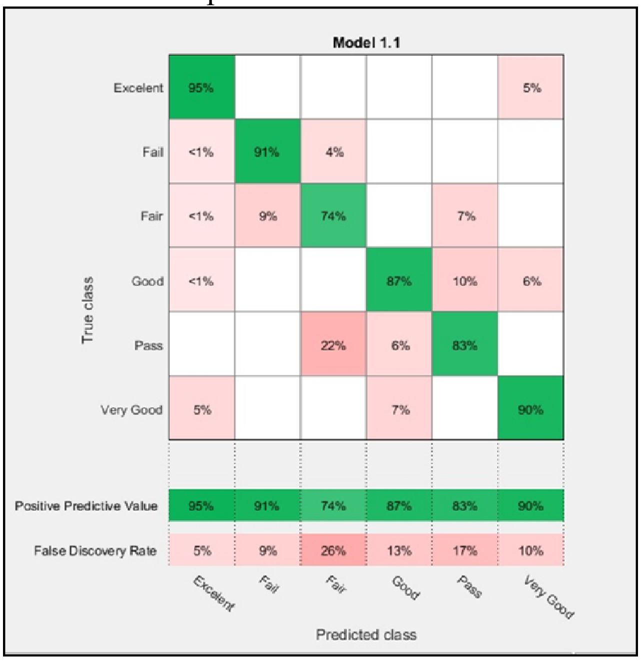

Fig.3: Fine KNN Confusion Matrix

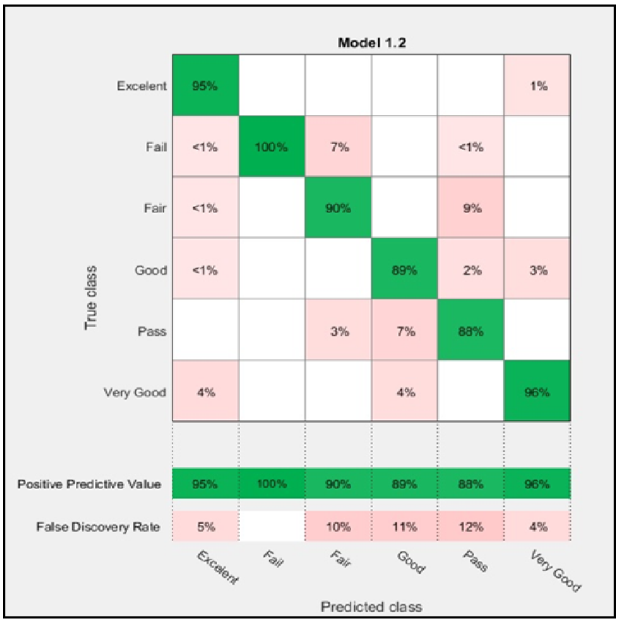

Fig.4: Medium KNN Confusion Matrix

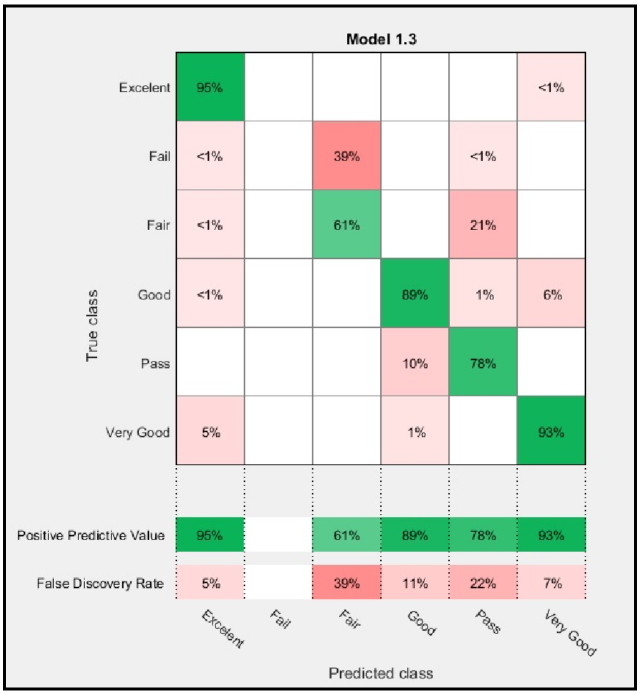

Fig.5: Coarse KNN Confusion Matrix

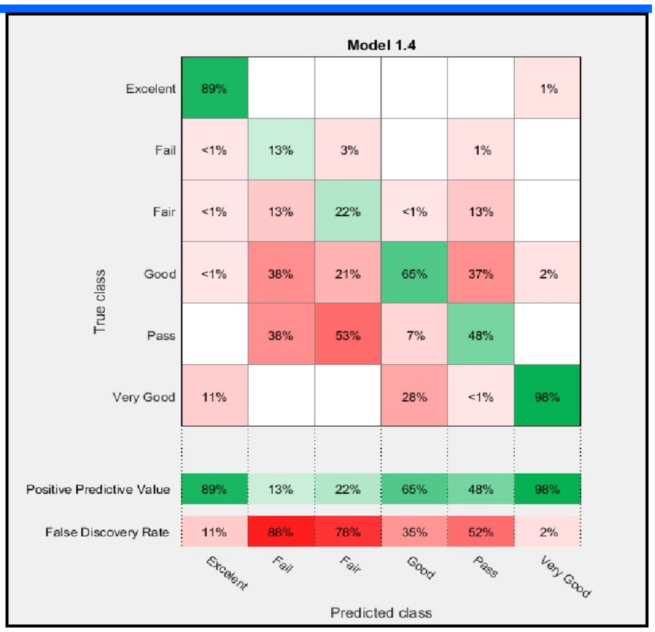

Fig.6: Cosine KNN Confusion Matrix

The Cosine KNN model has a large amount of false prediction shown in Fig.6. It gives 88% false prediction of Fail class and 98% positive prediction of Very Good class.

The Cubic confusion matrix in Fig.7 represents the positive and false prediction of classes according to the dataset. The best prediction value gives by the model is in Fail class with 100% positive prediction 13% of Pass class as false prediction

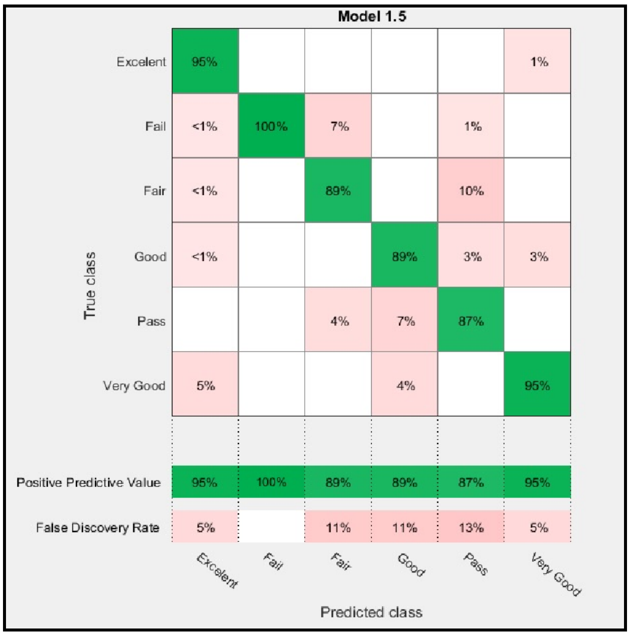

Fig.7: Cubic Confusion Matrix

he Weighted KNN model uses Euclidian distance metric to predict the nearest values according to the dataset. As in Fig.8 the confusion matrix of the positive and false prediction values shows that the best positive prediction is in Excellent class with 96% and the worst false prediction is in Pass class with 12%.

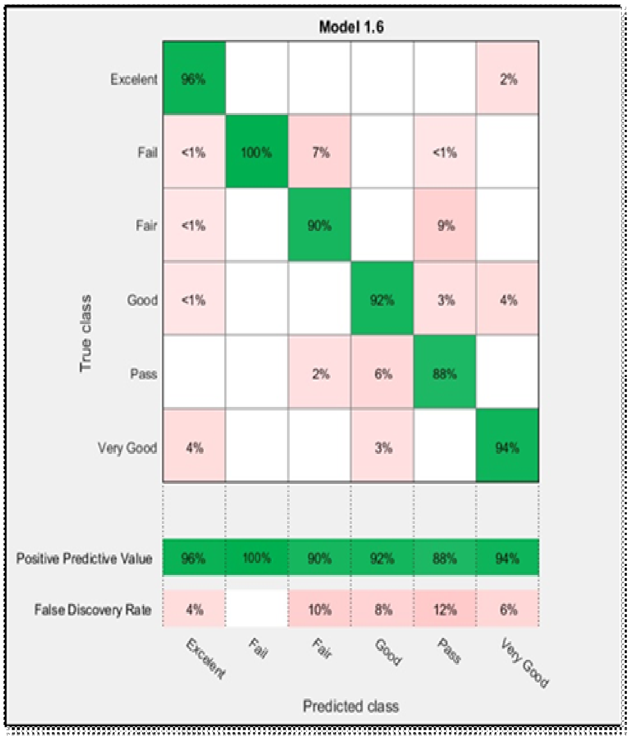

Fig.8: Weighted KNN Confusion Matrix

The prediction Graph shown in Fig.9 represents the classification of the classes according to the model applied on the dataset.

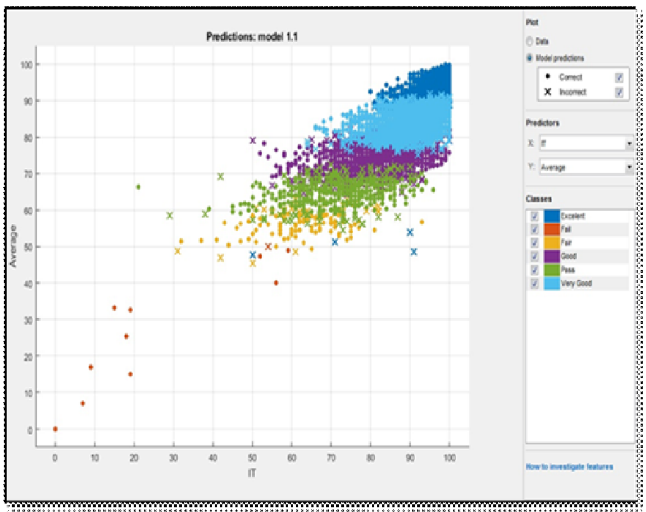

Fig.9: Prediction Graph

According to the results obtained from the conducted experiments, we found out that the most accurate algorithm used in classification is the Weighted KNN algorithm. It gives the best prediction accuracy according to the confusion matrix and the prediction graph. The Weighted KNN algorithm scored 94.1% accuracy and takes less time than other algorithms with 0.97885 Sec. On the other hand, Cosine KNN had the worst classification performance in which it scored 81.3% accuracy. However, the worst classifier in term of training time was the Cubic KNN with 30.441 Sec. The prediction model shows the predicted grade according to selecting of two attributes and creates the coordinator graph putting the selected attributes in X and Y axis, then show the predicted grade with correct and incorrect predictions .

5.0 Conclusion and future work

This study introduced a classification model for predicting the student's performance according to the marks of two subjects. We employed modified KNN classifiers such as Cosine KNN, Cubic KNN, and Weighted KNN for the classification. The dataset has been collected from Gaza Strip secondary schools, which has grade records for the students of the 11th scientific branch. Due to the classification experiment, we have shown that every algorithm gives a different value of accuracy and predictions due to the distance metrics used. The distance metrics, distance weight and number of neighbors are changeable variables that lead to enhance the efficiency of the algorithm results by increasing the accuracy of classification. On the other hand, the early prediction of the student's grade can help the principals to take decisions in order to help schools to initially identify students with low academic achievement and find ways to support them.

In the future, we will try to compare between the KNN classification algorithm and a different type of classification algorithm such as Naive Bayes algorithm to show the best in classification.

REFERENCES

1. Ahmad, F., N. H. Ismail and A. A. Aziz (2015). "The Prediction of Students' Academic Performance Using Classification Data Mining Techniques." Applied Mathematical Sciences 9(129): 6415-6426.
2. Asif, R., A. Merceron and M. K. Pathan (2014). "Predicting student academic performance at degree level: a case study." International Journal of Intelligent Systems and Applications 7(1): 49.
3. Baradwaj, B. K. and S. Pal (2012). "Mining educational data to analyze students' performance." arXiv preprint arXiv:1201.3417.
4. Bhatia, N. (2010). "Survey of nearest neighbor techniques." arXiv preprint arXiv:1007.0085.
5. Cover, T. and P. Hart (1967). "Nearest neighbor pattern classification." IEEE transactions on information theory 13(1): 21-27.
6. Deshpande, A., P. Pimpare, S. Bhujbal, A. Kommwar and J. Wagh (2016). "Student Performance Analysis, Visualization and Prediction Using Data Mining Techniques." Imperial Journal of Interdisciplinary Research 2(5).
7. Jeng-Shyang, P., Q. Yu-Long and S. Sheng-He (2004). "A fast K nearest neighbors classification algorithm." IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences 87(4): 961-963.
8. Jóźwik, A. (1983). "A learning scheme for a fuzzy k-NN rule." Pattern Recognition Letters 1(5-6): 287-289.

18.

9. Kononenko, I. and M. Kukar (2007). Machine learning and data mining: introduction to principles and algorithms, Horwood Publishing.
10. Mahapatra, R. P. and P. S. Chakraborty (2015). "Comparative Analysis of Nearest Neighbor Query Processing Techniques." Procedia Computer Science 57: 1289-1298.
11. Muja, M. and D. G. Lowe (2014). "Scalable nearest neighbor algorithms for high dimensional data." IEEE Transactions on Pattern Analysis and Machine Intelligence 36(11): 2227-2240.
12. Osmanbegović, E. and M. Suljić (2012). "Data mining approach for predicting student performance." Economic Review 10(1).
13. Papadopoulos, A. N. (2006). Nearest Neighbor Search:: A Database Perspective, Springer Science & Business Media.
14. Shakhnarovich, G., P. Indyk and T. Darrell (2006). Nearest-neighbor methods in learning and vision: theory and practice.
15. Wu, X., V. Kumar, J. R. Quinlan, J. Ghosh, Q. Yang, H. Motoda, G. J. McLachlan, A. Ng, B. Liu and S. Y. Philip (2008). "Top 10 algorithms in data mining." Knowledge and information systems 14(1): 1-37.
16. Yadav, S. K., B. Bharadwaj and S. Pal (2012). "Data mining applications: A comparative study for predicting student's performance." arXiv preprint arXiv:1202.4815.
17. Zhang, M.-L. and Z.-H. Zhou (2005). A k-nearest neighbor based algorithm for multi-label classification. 2005 IEEE international conference on granular computing, IEEE.